Self-Supervised Hyperspectral Learning for Leaf Multitrait Prediction With Sparse Labels

Chenye Su, Liangsheng Shi, Lijun Wang, Tinghan Wang, Yujie Wang, Jinmin Li, and Leilei He

***Abstract*—Accurate estimation of leaf traits is critical for evaluating ecosystem functions and elucidating vegetation–environment interactions. Hyperspectral remote sensing provides a non-destructive and information-rich means for trait analysis, yet its potential remains constrained by the scarcity of labeled data and the limited generalizability of existing learning methods. To overcome these challenges, we propose the spectral joint embedding predictive architecture (S-JEPA), a self-supervised learning (SSL) framework designed to learn generalizable spectral features by predicting target-region representations from contextual spectral information. Using the largest unlabeled hyperspectral dataset to date, comprising nearly 100 000 samples, S-JEPA enables accurate trait estimation with minimal labeled data. Across six representative leaf traits, S-JEPA achieved an average R^2 of 0.784, substantially exceeding conventional supervised, transfer-learning, and multitask learning models, and also surpassing another self-supervised baseline, spectral masked autoencoder (S-MAE), by a small margin. Furthermore, S-JEPA demonstrates superior robustness, especially in data-scarce scenarios, efficiently tackling a longstanding field challenge. The learned spectral features revealed coherent inter-trait associations aligned with known physiological linkages, demonstrating that S-JEPA captures biologically meaningful structures rather than statistical artifacts. This study highlights the efficacy of SSL in extracting generalizable spectral features and establishes a scalable framework for multitrait estimation and ecosystem monitoring.**

***Index Terms*—Data sparsity, deep learning, feature extraction, hyperspectral remote sensing, leaf trait, self-supervised learning (SSL).**

I. Introduction

LEAF traits represent critical indicators of plant physiological status and ecological function, reflecting their contribution to carbon and water cycling [1]. As key manifestations of the efficiency and regulation of core metabolic processes, including photosynthesis and transpiration [2], [3], these diverse traits exhibit tight coupling and jointly mediate plant function. For instance, carotenoids (CAR) regulate photoprotection [4], [5] and interact synergistically with chlorophyll (CHL) [6], collectively maintaining photosynthetic performance under stress. Similarly, structural traits like dry matter (DM) content influence leaf architecture and water retention [7], while leaf water content regulates photosynthetic vigor, a prerequisite for biomass accumulation [8]. The complex interrelationships among these traits are further modulated by species-specific physiology and environmental dynamics, demonstrating adaptive strategies of resource allocation under varying conditions [9], [10], [11]. These insights emphasize the necessity of holistic, multitrait analysis for understanding the intricate interactions between plants and the environment.

Traditional chemical analysis methods are limited by cumbersome sampling and measurement procedures, making it difficult to identify multiple traits simultaneously. Hyperspectral remote sensing has emerged as a nondestructive and high-throughput alternative [12], [13]. Using its high spectral resolution, hyperspectral data can sensitively detect specific signals across continuous narrow bands. These signals are closely related to the chemical and structural attributes of leaf tissues, hinting at microscopic physical processes like electronic transitions and chemical bond changes [14], [15]. This characteristic makes it possible to estimate leaf traits by interpreting spectral features. Interpretation methods typically fall into two categories: physical-based methods and empirical methods.

Physics-based methods, represented by radiation transfer models (RTMs), simulate the interaction between electromagnetic radiation and leaf tissues. Pioneering RTMs idealized leaves as multilayer rough absorption plates, driven by biochemical components like CHL content and structural parameters such as the number of leaf layers [16]. Subsequent improvements introduced more biochemical variables [17], [18] and more detailed leaf processes [19], [20]. However, RTMs consistently face the issue of equifinality, where different parameter sets could yield identical results [21]. Furthermore, a large number of traits and processes remain unaccounted for in these models [22], [23], which constrains their accuracy and applicability under real-world conditions.

Empirical methods, conversely, establish connections between traits and spectra by fitting observational data rather than deriving from physical laws. Initial efforts included

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The code and trained model objects are available online at https://github.com/CyanKirin99/sjepa.

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spectral transformation [24], [25], band selection [26], and vegetation indices [27], [28]. In recent years, the adoption of machine learning and deep learning [29], [30], [31], [32], [33] has significantly enhanced the models' ability for autonomous feature extraction, thereby improving prediction performance. To address the limitation that these methods are often restricted to specific traits and controlled conditions, approaches like transfer learning (TL) and multitask learning (MTL) have been explored to improve model generalizability. Fine-tuning is the primary strategy in TL [34], using data from different conditions or traits as source tasks to provide prior knowledge for new traits or conditions designated as target tasks [35], [36], [37]. MTL strategies simultaneously optimize the loss functions for multiple traits, thereby extracting common shared features across these traits [38].

In spite of notable progress, current methodologies still face persistent limitations in simultaneously estimating multiple traits. A predominant limitation is their dependence on labeled datasets. The methods mentioned above all fall under the category of supervised learning, which updates model parameters by minimizing the difference between predictions and trait labels [39]. However, the scarcity of such labels severely undermines the effectiveness of this paradigm, as insufficient supervision often prevents models from learning robust features and leads to overfitting or even training instability [40]. Another major problem of the supervised learning paradigm lies in its inherent bias in feature extraction toward the characteristics and distribution of the training labels. The success of TL largely depends on the degree of alignment between source and target tasks. During source-task optimization, the model tends to emphasize spectral features specific to the associated traits or conditions. When such features are transferred to target tasks with spurious or nonexistent correlations, they may not only fail to enhance but even degrade predictive performance [41].

Consequently, reducing the dependency on labeled data while enhancing the extraction of generalizable spectral features has become a pivotal challenge in advancing multitrait estimation. Addressing this challenge requires two essential components: a large-scale hyperspectral database encompassing sufficient and diverse samples to support generalizable feature learning, and a learning algorithm capable of robust representation learning even in the absence of supervision. In recent years, collective efforts from the international research community have led to the development of large-scale hyperspectral databases [42]. Although associated trait labels remain sparse, the spectra themselves—with broad geographic, species, and sensor variability—provide a valuable foundation for learning generalizable features. On the algorithmic front, self-supervised learning (SSL) has emerged as a transformative method. By constructing appropriate pretext tasks, SSL enables the discovery of inherent structural patterns within data, deftly avoiding the reliance on explicit labels [43]. The effectiveness of SSL has been well-demonstrated in fields like computer vision [44], [45] and natural language processing [46], [47]. Its potential is also increasingly recognized in hyperspectral analysis, where it shows promise across a range of tasks [48], [49], [50].

This research presents the following contributions.

1) We propose the spectral joint-embedding predictive architecture (S-JEPA), a SSL paradigm trained on the largest leaf hyperspectral dataset to date, comprising nearly 100 000 samples across diverse global species and ecosystems. This framework extracts generalizable spectral features and translates them into accurate multitrait estimations via a cross-attention mechanism.
2) We conducted rigorous comparative experiments across multiple traits and gradients of label availability. Alongside existing methods—partial least squares regression (PLSR), End-to-end deep learning (End-to-end), TL, and MTL—we also implemented an additional self-supervised baseline (Spectral Masked Autoencoder, S-MAE) for comparison, which consistently demonstrated S-JEPA's superior performance.
3) We explored the mechanisms and reasons behind S-JEPA to robustly estimate multiple traits, and demonstrated its ability to capture multitrait relationships. A multitrait node network, constructed from the dense prediction matrix generated by S-JEPA, was further used to examine how the model behavior reflects the complex real-world relationships among traits from a unified spectral perspective.

II. Materials and Methods

A. Data Acquisition and Processing

We compiled a large dataset of leaf spectra and corresponding physiological traits from 69 projects, including experimental campaigns and publicly accessible repositories (see Fig. A.1, Table A.1 in the Appendix A of the supplementary material). This dataset spans diverse geographical regions, plant species, and ecological contexts, providing a solid foundation for developing generalizable models.

1) Hyperspectral Data: Hyperspectral data in the dataset were collected using various instruments (e.g., ASD FieldSpec 4), each with distinct spectral resolutions and wavelength spans. To ensure consistency, raw spectral signatures were processed using Savitzky–Golay filtering [51] to smooth the spectra and reduce noise. These data were then standardized by interpolating all spectra to 1 nm intervals within the 350–2500 nm range, resulting in 99 303 uniformly resampled spectral samples.

The 1st–99th percentile quantiles profiles closely align with the mean spectrum, indicating strong consistency across the dataset (see Fig. 1). Reflectance is generally low in the visible region, with a peak near 550 nm, and rises sharply around the red-edge (~700 nm), then gradually declines in the shortwave infrared, except for pronounced absorption features at 1500 nm and 1900 nm. The coefficient of variation (CV) is elevated in the visible range, near the shortwave infrared absorption troughs, and at both spectral ends, suggesting richer information content in these regions.

2) Trait Data: We selected six key leaf traits: leaf mass per area (LMA, g/m^2), water content (Water, %DM), CHL content (CHL, %DM), CARs, %DM), nitrogen content (Nitrogen, %DM), and carbon content (Carbon, %DM). To ensure consistency, all concentration-related traits were expressed on a mass

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TABLE I
METHODOLOGICAL DISTINCTIONS BETWEEN PARADIGMS OF SUPERVISED LEARNING AND SSL

Methodological dimension	Supervised learning	Self-supervised learning
Operational logic	Explicit mapping: $f(X) \rightarrow Y$.	Latent representation: $Z = Enc(X)$
Supervision source	Extrinsic: Relies on expensive laboratory trait measurements.	Intrinsic: Relies on the internal correlation between spectral bands.
Data scope	Restricted: Limited to small, manually labeled subsets.	Scalable: Leverages massive global datasets without labels.
Feature utility	Static: Features are biased toward specific traits or conditions.	Generalizable: Features form a universal pool for diverse downstream tasks

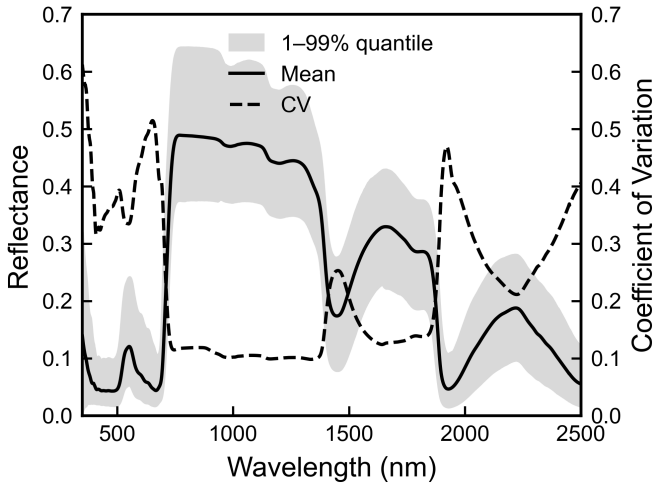


Fig. 1. Distribution and CV of leaf spectral reflectance across wavelengths.

basis. For traits originally measured per unit area (e.g., CHL in μg/cm^2), values were converted to mass-based units (i.e., %DM) by dividing by LMA. Outliers for each trait were removed using the 3σ rule, and the values were subsequently normalized using a robust scaler [52] to minimize the influence of extremes.

The above processing resulted in a total of 43 421 samples with at least one trait label (see Fig. 2). Because of the sparsity of labels, the number of valid samples varies substantially across different traits. Among the six traits, LMA exhibited the largest number of valid samples, totaling 42 687, while the remaining five traits ranged between 2000 and 6000 samples. Moreover, the distributions of these traits differ considerably. CHL and CAR approximate a normal distribution, whereas LMA, water, and nitrogen exhibit varying degrees of right skewness. In contrast, Carbon displays a distinct bimodal distribution with a slight left skew. Further details are provided in Table A.2 in Appendix A of the supplementary material.

The full collection of 99 303 spectral samples (irrespective of trait labels) was used as the large unlabeled dataset for the self-supervised pretraining phase of the S-JEPA model (see Section II-B1). Subsequently, the 43 421 samples with trait labels were employed for training and evaluating the downstream modules for leaf trait estimation (see Section II-B2).

B. Model Architecture and Training Strategies

In this study, we introduced an SSL paradigm to leaf trait estimation for the first time. The core innovation lies in shifting the learning objective from the traditional explicit mapping to capturing the intrinsic structural properties of the spectral data itself. As summarized in Table I, this shift fundamentally redefines the supervision mechanism and data use scale, by decoupling feature extraction from label dependency, the proposed framework enables the model to learn from massive unlabeled datasets, effectively surmounting the constraints of label sparsity that hinder conventional supervised methods.

1) SSL via S-JEPA: Harnessing SSL on a large hyperspectral dataset, we developed a framework proficient in extracting generalizable spectral features. This approach draws inspiration from the I-JEPA framework, which learns latent representations by predicting target region features from contextual counterparts within a high-dimensional space [44]. Representations in the high-dimensional latent space offer superior informational capacity over raw reflectance values, encompassing the complex and interactive relationships among spectral bands inherent within the hyperspectral data. This provides a solid foundation for estimating multiple traits.

To exploit the unique properties of hyperspectral data, we developed the spectral joint embedding predictive architecture (S-JEPA), comprising three integral modules:

a) Embedder: This module transforms raw hyperspectral data into embedding vectors. Adapting techniques from computer vision [53], we segment the spectra into fixed-length patches, projecting each into a high-dimensional embedding space via linear transformations. To augment the informativeness of these representations, first- and second-order spectral derivatives are computed, projected, and concatenated with the original embeddings along the feature dimension. A learnable positional encoding mechanism is incorporated to preserve the sequential integrity of spectral bands.

b) Encoder: The Encoder maps embedding vectors into high-dimensional feature representations, employing

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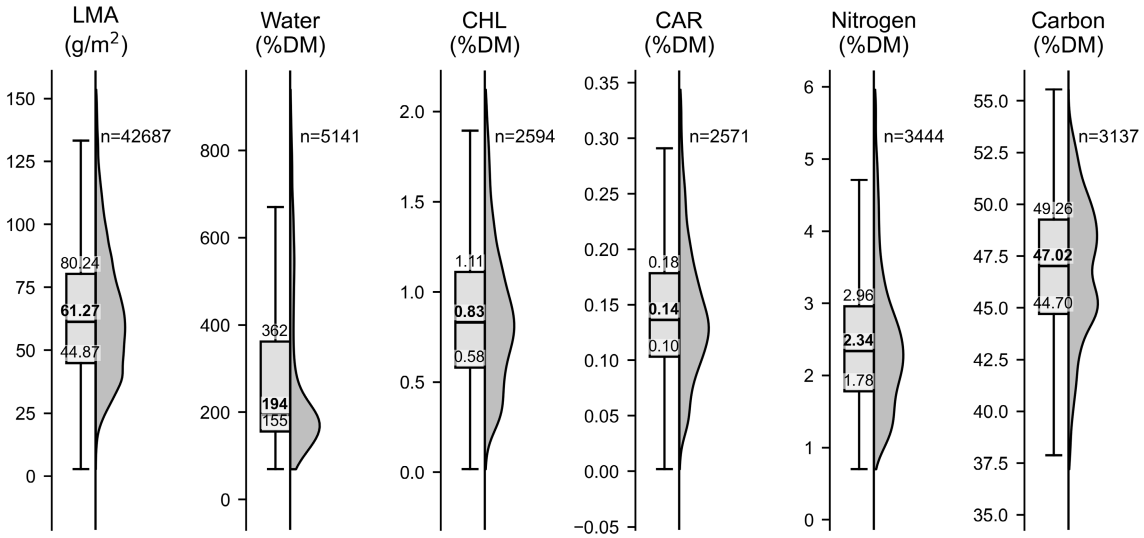


Fig. 2. Distribution of six leaf traits. Each subplot comprises (left) a boxplot depicting the median and interquartile range, and (right) a kernel density estimate (KDE) illustrating the probability density of trait values.

a standard Transformer Encoder [54] with multihead self-attention to capture global dependencies. The complex interband correlations and high-dimensional nature of hyperspectral data challenge traditional models; Transformers, however, excel in discerning long-range interactions, making them well-suited for this task [55]. Their adaptability ensures robust performance across heterogeneous ecological datasets.

c) Predictor: Tasked with predicting target region feature representations from contextual inputs, the Predictor is designed as a lightweight Transformer with reduced layers and embedding dimensionality relative to the Encoder. This lightweight design constrains its capacity to extract features independently, thereby encouraging the use of contextual information and reinforcing the Encoder's role in learning effective representations.

The training protocol unfolds in five phases (see Fig. 3).

1) *Spectral Partitioning:* Randomly delineate nonoverlapping regions from hyperspectral band patches, assigning context and target roles.
2) *Spectral Vectorization:* The Embedder converts raw data into embedding vectors.
3) *Feature Encoding:* The Encoder processes these vectors to yield feature representations.
4) *Latent Prediction:* The Predictor predicts target region features using contextual representations and positional data.
5) *Parameter Optimization:* The disparity between predicted and actual target features informs the loss function, driving parameter optimization.

To prevent representation collapse, we employed dual Encoders—structurally identical but parametrically distinct—for context and target regions, respectively. The Context Encoder and the predictor are updated via backpropagation, while the Target Encoder adopts an exponential moving average (EMA) for asynchronous alignment. During training, the two Encoders gradually converge toward consistent representations. Training efficiency is enhanced by sampling one context region and multiple target regions per sample (see Algorithm B.1 in Appendix B of the supplementary material).

2) Translating Generalizable Spectral Features Into Trait Predictions: We developed a lightweight decoder module, driven by a cross-attention mechanism, to translate the learned generalizable spectral features into specific trait predictions. The holistic inference model for trait prediction (see Fig. 4) consists of dedicated Decoders for each trait, integrated with the pretrained Embedder and Encoder from S-JEPA.

The Decoder comprises two key components: a learnable trait-specific token (T_{TS}) and a trait-specific multilayer perceptron ($\mathrm{MLP}_{\mathrm{TS}}$). T_{TS} interacts with the extracted spectral features through a cross-attention mechanism to capture trait-relevant information, while $\mathrm{MLP}_{\mathrm{TS}}$ refines these features for final prediction via two fully connected layers with a gaussian error linear unit (GELU) activation inserted between them.

Generally, cross-attention enables the interaction between two distinct sets of representations by allowing one set (queries, Q) to selectively attend to another set (keys, K and values, V). Attention weights, computed based on the similarity between Q and K, are used to aggregate information from V [54]. In our implementation, the trait-specific token (T_{TS}) acts as the query (Q), and the generalizable spectral features (F) serve as both the key (K) and value (V). This design allows the T_{TS} for each trait to aggregate trait-specific features (F_{TS}) from the generalizable spectral features (F), formalized as

$$F_{\mathrm{TS}} = \mathrm{softmax}\left(\frac{T_{\mathrm{TS}}F^{\mathrm{T}}}{\sqrt{d_k}}\right)F \tag{1}$$

where d_k represents the dimensionality of the feature representations. The aggregated trait-specific features (F_{TS}) are then refined by the corresponding $\mathrm{MLP}_{\mathrm{TS}}$, which performs dimensionality reduction and outputs the final trait predictions ($\hat{y}$). Further implementation details are provided in Algorithm B.2 in Appendix B of the supplementary material.

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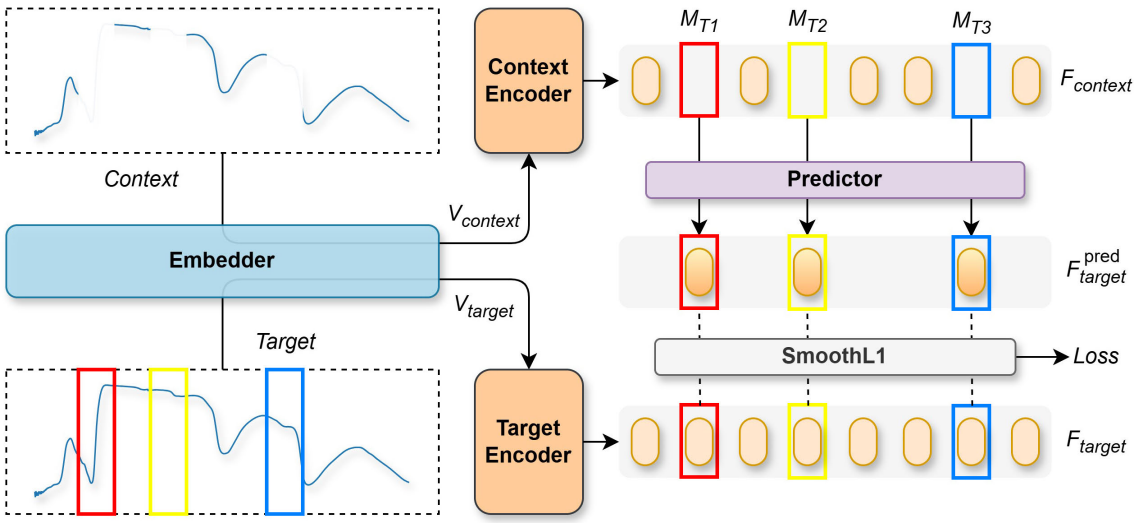


Fig. 3. Schematic of the SSL workflow in S-JEPA.

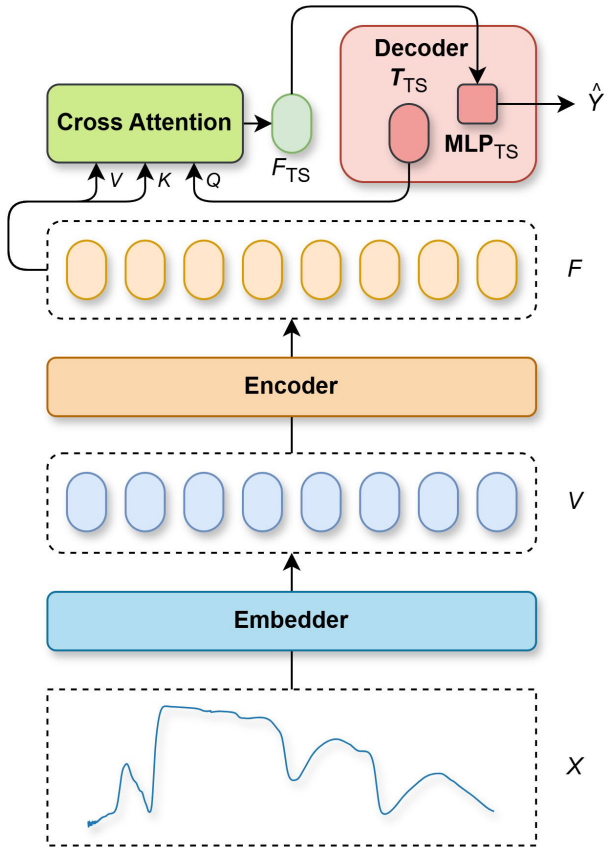


Fig. 4. Unified inference architecture shared by all deep learning-based prediction models in this study, including End-to-end, TL, MTL, S-MAE, and S-JEPA.

A dedicated Decoder for each trait is trained using the available labeled data, while they can be jointly deployed during inference since all Decoders share a common Embedder and Encoder. This significantly reduces parameter redundancy and improves efficiency.

3) Competitive Models: To rigorously evaluate S-JEPA, we benchmarked it against both PLSR as a traditional baseline and against deep learning variants that share its architecture but differ in training paradigms: End-to-end, TL, and MTL. This architectural consistency among the deep-learning models ensures that any observed performance differences between them can be primarily attributed to the variations in training strategies.

a) Partial least squares regression: This classic method projects data into a lower-dimensional space, maximizing covariance between spectral features and traits [56]. We set 200 components to balance representational fidelity and computational feasibility (see Algorithm B.3 in Appendix B of the supplementary material).

b) End-to-end: This strategy adheres to the conventional paradigm of deep neural network training. All model parameters—including those of the embedder, encoder, and decoder—were jointly optimized using paired hyperspectral data and trait labels (see Algorithm B.4 in Appendix B of the supplementary material). End-to-end serves as a benchmark for evaluating the model's intrinsic learning capacity.

c) Transfer learning: TL leverages features learned from a source trait for a target trait. The Embedder and Encoder are first trained End-to-end using source-trait labels, followed by decoder fine-tuning with target-trait labels (see Algorithm B.5 in Appendix B of the supplementary material). Source-target permutations across six traits were executed.

d) Multitask learning: Joint optimization of multiple trait-specific loss functions promotes shared representations and mitigates overfitting. The model employs a shared embedder-encoder architecture with parallel trait-specific decoders, enabling concurrent multitrait prediction (see Algorithm B.6 in Appendix B of the supplementary material).

e) S-MAE: This strategy learns spectral representations by reconstructing masked wavelength segments from visible regions. A lightweight Decoder is employed during pretraining to recover the original spectral sequence from the outputs of the embedder and encoder. In contrast to S-JEPA, it focuses on restoring fine-grained spectral details rather than modeling

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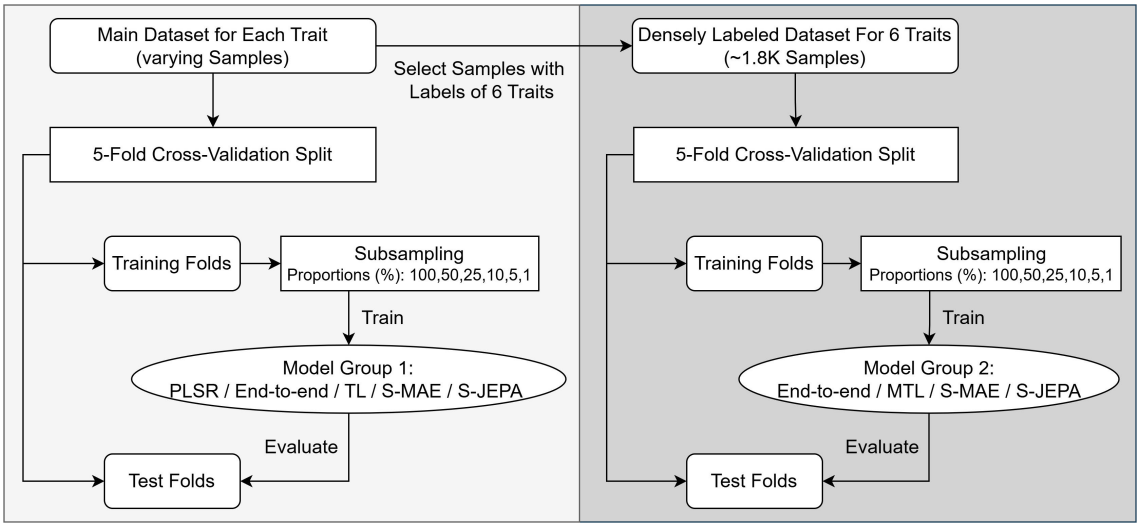


Fig. 5. Schematic of comparative experiments for predictive performance.

high-dimensional latent relationships (see Algorithm B.7 in Appendix B of the supplementary material).

4) Training Conditions and Setup: Spectral segmentation was performed using a 30 nm window. The Embedder and Encoder had feature dimensions of 768, while the predictor in S-JEPA and the decoder in S-MAE each had 96 dimensions. Both models adopted identical spectral masking configurations, consisting of four contiguous 6-unit target blocks and one noncontiguous 24-unit context block.

Model optimization employed the AdamW optimizer with linear warm-up and cosine annealing schedules (learning rates: initial = 2×10^{-5}, peak = 1×10^{-4}, final = 1×10^{-6}). Both S-JEPA and S-MAE were pretrained for 1000 epochs. During downstream training, only the trait-specific decoders were updated for S-JEPA, S-MAE, and TL models, whereas the End-to-end and MTL models optimized all parameters jointly. The single-trait models were trained for 500 epochs per trait, while the MTL models were trained for 5000 epochs until convergence.

All computations were performed using Python 3.12, PyTorch 2.5.1, and scikit-learn 1.6.1 on a workstation equipped with an Intel i7-14700KF CPU and an NVIDIA RTX 4080 GPU.

C. Comparative Experiments for Predictive Performance

Model performance was assessed by conducting all comparative experiments on six traits (see Fig. 5). For each trait, available labeled data (see Table A.2 in Appendix A of the supplementary material) were independently used to train and evaluate the performance of the first group of models: PLSR, End-to-end, TL, S-MAE, and S-JEPA. Given that the MTL model requires labels for all six traits for training, we selected densely annotated samples from the dataset (see Table A.3 in Appendix A of the supplementary material). This subset was then used to evaluate the performance of the second group of models: End-to-end, MTL, S-MAE, and S-JEPA.

To ensure the reliability of results, we performed fivefold cross-validation on both sets of experiments to partition the data into training and test sets. In each fold, models were first trained using the entire set of available labeled training data (100% proportion) to serve as a reference under current label availability. To simulate scenarios with limited labeled data, subset sampling was then applied to the training set at proportions (50%, 25%, 10%, 5%, and 1%). All subset sampling procedures were performed using a fixed random seed to ensure comparability across different models.

The coefficient of determination (R^2), normalized root-mean-square error (nRMSE), and R^2 standard deviation (R^2 SD) were used as performance metrics for evaluation.

D. Analysis of Spectral Features

Attention weights in the cross-attention mechanism reflect spectral band importance for trait prediction, enhancing model interpretability. In this study, we examined the Decoder-derived attention weights to assess the physical plausibility of the deep-learning models. For each trait, attention weights were averaged across samples and attention-heads, normalized between their max-min range, and aggregated into a global attention distribution to quantify band-specific contributions. This distribution reflects each spectral band's relative importance, revealing the models' spectrum-trait relationships.

To further elucidate the model's decision-making process, we additionally employed the integrated gradients (IGs) method [57] to quantify band-level contributions at the sample level. For each trait, three representative samples (high, median, and low values) were selected from the same dataset to control variability. The spectral profile of the median sample served as the baseline, and IG was computed as the path integral of the input gradients along the linear interpolation between the baseline and the target sample

$$IG_i(x) = \left(x_i - x_i'\right) \int_0^1 \frac{\partial F\left(x' + \alpha\left(x - x'\right)\right)}{\partial x_i} d\alpha \tag{2}$$

where x and x' denote the spectral vectors of the target and baseline samples, respectively, and F represents the model output. This analysis enables localized interpretation of

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model predictions and complements the global attention-based analysis.

E. Validation and Network Analysis of Multitrait Relationships

To assess whether S-JEPA effectively preserved and generalized the intrinsic inter-trait dependencies, we conducted a two-stage analysis combining quantitative validation and network-based exploration.

1) Validation of Multitrait Relationship Consistency: We first evaluated whether the six benchmark traits predicted by the independently fine-tuned S-JEPA models retained the statistical dependencies observed among the ground-truth measurements. The similarity between the predicted and observed joint trait distributions was quantified using the Jensen–Shannon (JS) divergence, defined as

$$JS(P,Q) = \frac{1}{2}\left[KL(P \parallel M) + KL(Q \parallel M)\right]$$
$$M = \frac{1}{2}(P+Q) \tag{3}$$

where P and Q denote the empirical and predicted joint distributions, and $KL(P \parallel M)$ is the Kullback–Leibler divergence of P with respect to M. This metric was applied across varying training set proportions to assess how well the joint distributions from model predictions aligned with those from observational data. Lower JS values indicate a closer match between the predicted and observed multitrait structures.

In addition, pairwise Pearson correlation coefficients were computed for both predicted and observed traits to examine the linear consistency among traits. To mitigate indirect effects mediated by other variables, partial correlations were also estimated as

$$\rho_{ab\cdot c} = \frac{\rho_{ab} - \rho_{ac}\rho_{bc}}{\sqrt{\left(1-\rho_{ab}^2\right)\left(1-\rho_{bc}^2\right)}} \tag{4}$$

where ρ_{ab} is the Pearson correlation between traits a and b, with similar interpretations for other terms. The congruence between the predicted and observed correlation matrices was further tested using the Mantel statistic, defined as

$$r_M = \frac{\sum_{i<j}\left(A_{ij}-\bar{A}\right)\left(B_{ij}-\bar{B}\right)}{\sqrt{\sum_{i<j}\left(A_{ij}-\bar{A}\right)^2 \sum_{i<j}\left(B_{ij}-\bar{B}\right)^2}} \tag{5}$$

where A and B are the upper-triangular elements of the observed and predicted correlation matrices, and $\bar{A}$ and $\bar{B}$ are their respective means. Statistical significance was determined using a permutation test.

2) Network Construction Based on Dense Trait Predictions: Building upon these validated consistencies, we further extended the analysis to a broader trait set. The pretrained S-JEPA embedder and encoder were used to predict twelve traits, including the six benchmark traits and six additional traits with limited labels (see Table A.4 in Appendix A of the supplementary material). For each additional trait, a lightweight decoder was trained on top of the frozen S-JEPA representation, enabling prediction across approximately 43 000 hyperspectral samples with at least one labeled trait. These predictions collectively formed a dense multitrait matrix, from which a trait-level node network was derived using the Python library networkx.

In this network, each node represents a trait, and the edges denote pairwise Pearson correlations among predicted values. To account for the uncertainty of correlation estimation, we employed a bootstrapping procedure (200 resamples) to generate the distribution of each edge weight. The mean correlation represents the association strength, while the SD across bootstraps reflects its reliability, which was visualized as edge thickness in the network graph.

III. Results

A. Predictive Performances of Models for Multiple Traits

1) Scenarios With Full Labeled Training Sets: The S-JEPA model demonstrated superior performance among all evaluated models (see Fig. 6), with a mean R^2 of 0.784 and a mean nRMSE of 0.075 across all six traits. The S-MAE model achieved overall performance closely matching that of S-JEPA, with a mean R^2 of 0.779 and nRMSE of 0.076. The End-to-end model outperformed the PLSR model, achieving higher accuracy (R^2 = 0.697 versus 0.655) and lower error (nRMSE = 0.090 versus 0.096). The performance of TL models depends on the correlation between source and target traits. In intra-trait transfer scenarios, TL models demonstrated performance comparable to the End-to-end model. For bidirectional inter-trait transfer between strongly correlated traits (e.g., CAR and CHL), the TL models' average R^2 of 0.703 was comparable to S-JEPA's R^2 of 0.737. However, for weakly correlated pairs (e.g., Carbon and CAR), the accuracy of TL declined obviously (R^2 = 0.412).

Model stability was further assessed using fivefold cross-validation (see Table C.1 in Appendix C of the supplementary material). Both the S-JEPA and S-MAE models demonstrated superior stability, evidenced by the lowest mean SD of R^2 (0.095 and 0.094, respectively) across the six traits during cross-validation. In contrast, PLSR, End-to-end, and TL models exhibited greater performance variability, with mean R^2 SDs of 0.139, 0.134, and 0.137, respectively.

Substantial variation in predictability was observed among the different traits. Water was highly predictable, with R^2 values ranging from 0.710 to 0.950 and nRMSE values from 0.023 to 0.057 across the evaluated models and scenarios. Conversely, CAR proved more challenging to predict, with observed R^2 values spanning 0.394–0.715 and nRMSE values from 0.096 to 0.141 under different modeling approaches. These disparities in predictive performance underscore the varying complexities of the relationships between hyperspectral signatures and individual traits.

2) Scenarios With Varying Labeled Training Sets: It is seen that model performance generally declined as the proportion of labeled training data diminished. The degradation followed a logarithmic pattern, with increased prediction variability and sharp drops at smaller training sizes [see Fig. 7(a)]. Across all traits and training proportions, the S-JEPA model achieved a mean R^2 of 0.580, while the S-MAE model attained a comparable value of 0.582, both substantially outperforming

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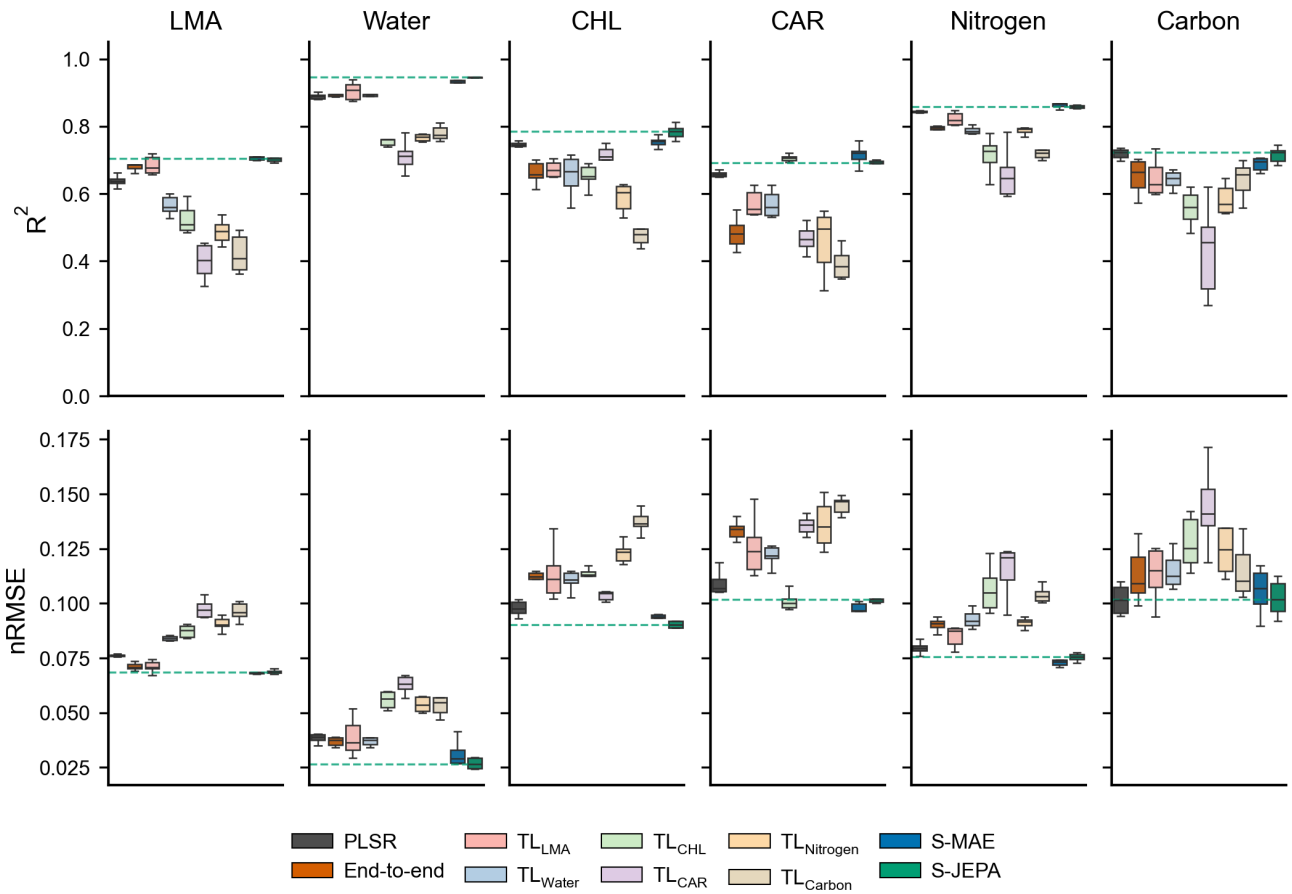


Fig. 6. Predictive performances (R^2 and nRMSE) of the PLSR, End-to-end, TL, S-MAE, and S-JEPA models across six traits.

the PLSR (−2.165), End-to-end (0.344), and TL (0.422) models. These results highlight the superior robustness of S-JEPA and S-MAE, which maintained R^2 values above 0.5 when more than 25% of labeled data were available. Notably, although S-MAE exhibited similar overall accuracy, it showed an earlier onset of performance degradation for specific traits such as Carbon and LMA. Among all evaluated models, PLSR showed the greatest susceptibility to data reduction, experiencing a significant deterioration in accuracy. The End-to-end model demonstrated intermediate data requirements compared to other approaches. The performance of TL models under a reduced training dataset demonstrated strong dependence on source-target domain relationships. Specifically, intra-trait transfers maintained stable performance when trained on only 5% of the full dataset. Strongly correlated trait pairs (e.g., CHL and CAR) maintained predictive accuracy ($R^2 > 0.5$) even at reduced training set sizes (10% of full data). However, for weakly associated traits, predictive accuracy deteriorated below acceptable thresholds at reduced training proportions.

Further analyses of performance stability [see Fig. 7(b)] provided additional insights into model behavior under data scarcity. As the volume of training data decreased, the SD of R^2 values generally increased, reaching its peak when only 1% of the labeled data was available. The S-JEPA (SD = 0.367) and TL models (SD = 0.377) exhibited lower R^2 variability compared to the End-to-end model (SD = 0.424). This contrasted sharply with the PLSR model, which displayed extreme R^2 variability (SD = 7.572), particularly at the lowest training data proportions.

Under the densely labeled dataset, both S-JEPA and S-MAE substantially outperformed MTL and End-to-end models (see Table II, Fig. C.1, Fig. C.2, and details in Table C.2 in Appendix C of the supplementary material). While S-MAE achieved the highest mean R^2 on the full training set (0.778 versus S-JEPA's 0.743), S-JEPA demonstrated superior robustness in data-scarce scenarios, surpassing S-MAE's performance from the 25% level downward and maintaining a strong R^2 of 0.517 even with only 5% of the labeled samples. The MTL model showed a clear advantage over the End-to-end model (e.g., R^2 0.483 versus 0.206 at 25%), but its performance degraded rapidly as labels decreased, collapsing alongside the End-to-end model at 10%.

B. Global and Local Spectral Interpretability of Models

The comparative analysis revealed distinct attention patterns among models trained with different learning strategies. The End-to-end model [see Fig. 8(a)] displayed a relatively uniform, smooth attention distribution across the entire spectral range. TL models [see Fig. 8(b) and (c)] showed consistent attention patterns among homologous models, with broader spectral activation when TL was applied to identical source and target traits.

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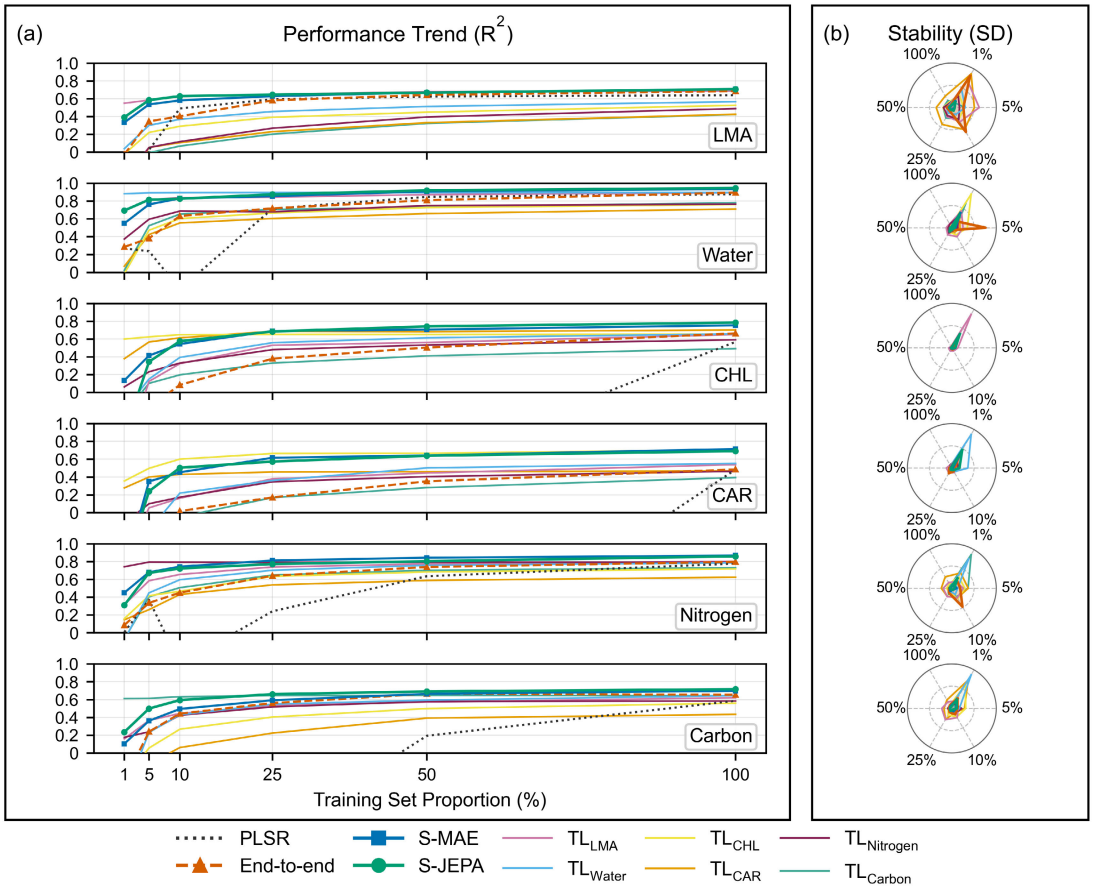


Fig. 7. (a) R^2 of the PLSR, End-to-end, TL, S-MAE, and S-JEPA models across training set proportions for six traits. (b) Radar plot of model R^2 SDs across traits and proportions, excluding PLSR due to its excessively high SD, to ensure clear model comparison.

By contrast, the S-MAE [see Fig. 8(d)] and S-JEPA [see Fig. 8(e)] models both presented discontinuous, localized attention patterns characterized by sparse high-response bands. Despite their structural similarity, their spectral emphases varied: for photosynthetic pigments (CHL and CAR) and related elemental traits (Nitrogen and Carbon), S-MAE primarily focused near 550 nm (green region), whereas S-JEPA additionally emphasized the red-edge region around 700 nm.

Further sample-level interpretability analysis based on IGs for nitrogen prediction (see Fig. 9) revealed that the 700 nm red-edge acts as a distinct spectral boundary separating high- and low-value samples, beyond which IG fluctuations diminished markedly. Similar IG analyses for five additional traits (see Appendix E of the supplementary material) confirmed that spectral regions prioritized in global attention maps often correspond to wavelengths exerting significant local influence at the sample level. Traits such as LMA exhibited more frequent and pronounced IG oscillations, suggesting higher spectral heterogeneity and sensitivity compared with other traits.

C. Statistical and Structural Characterization of S-JEPA-Predicted Multitrait Relationships

The statistical dependencies among multiple traits were consistently preserved in the S-JEPA predictions. Comparison between the joint distributions of predicted and observed traits showed that S-JEPA achieved markedly lower JS divergence (mean = 0.058) than the End-to-end model (mean = 0.093) across all training data proportions [see Fig. 10(a)], indicating a closer alignment with the empirical multitrait structure. Moreover, JS divergence remained stable under reduced training data, suggesting that S-JEPA maintained distributional consistency even with limited supervision.

Correlation analyses further confirmed the accurate reconstruction of inter-trait relationships. Both observed [see Fig. 10(b)] and the S-JEPA-predicted [see Fig. 10(c)] traits displayed comparable covariance patterns, where simple correlations were substantially stronger than partial correlations (mean absolute r: 0.527 versus 0.150). Strong positive associations were found among CHL, CAR, and Nitrogen, while Carbon exhibited a negative correlation with Nitrogen. After controlling for indirect effects, only the CHL–CAR linkage remained dominant. The S-JEPA predictions reproduced these structures with high fidelity, as supported by Mantel tests (simple: r = 0.996, p = 0.002; partial: r = 0.952, p = 0.001).

Extending this analysis to the dense twelve-trait prediction matrix, a multitrait node network was constructed (see Fig. 11). The pigments (CHL, CAR, CHLa, CHLb) and nitrogen formed a compact positive-correlation cluster, reflecting coordinated physiological variation among nitrogen and pigment-related traits. Water served as a structural hub, showing strong positive associations with potassium, phosphorus, and magnesium, whereas its correlation with carbon and LMA

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TABLE II
PREDICTIVE PERFORMANCE METRICS OF THE END-TO-END MTL, S-MAE, AND S-JEPA MODELS ON DENSELY LABELED DATASET FOR SIX TRAITS

Training set proportion	Model type	Mean R^2	Mean nRMSE
100	End-to-end	0.411	0.118
	MTL	0.702	0.082
	S-MAE	**0.778**	**0.071**
	S-JEPA	0.743	0.077
50	End-to-end	0.273	0.132
	MTL	0.591	0.097
	S-MAE	**0.743**	**0.077**
	S-JEPA	0.701	0.082
25	End-to-end	0.206	0.137
	MTL	0.483	0.110
	S-MAE	0.664	0.088
	S-JEPA	**0.669**	**0.087**
10	End-to-end	0.142	0.143
	MTL	0.146	0.141
	S-MAE	0.554	0.101
	S-JEPA	**0.611**	**0.095**
5	End-to-end	0.051	0.149
	MTL	-0.534	0.188
	S-MAE	0.458	0.110
	S-JEPA	**0.517**	**0.105**

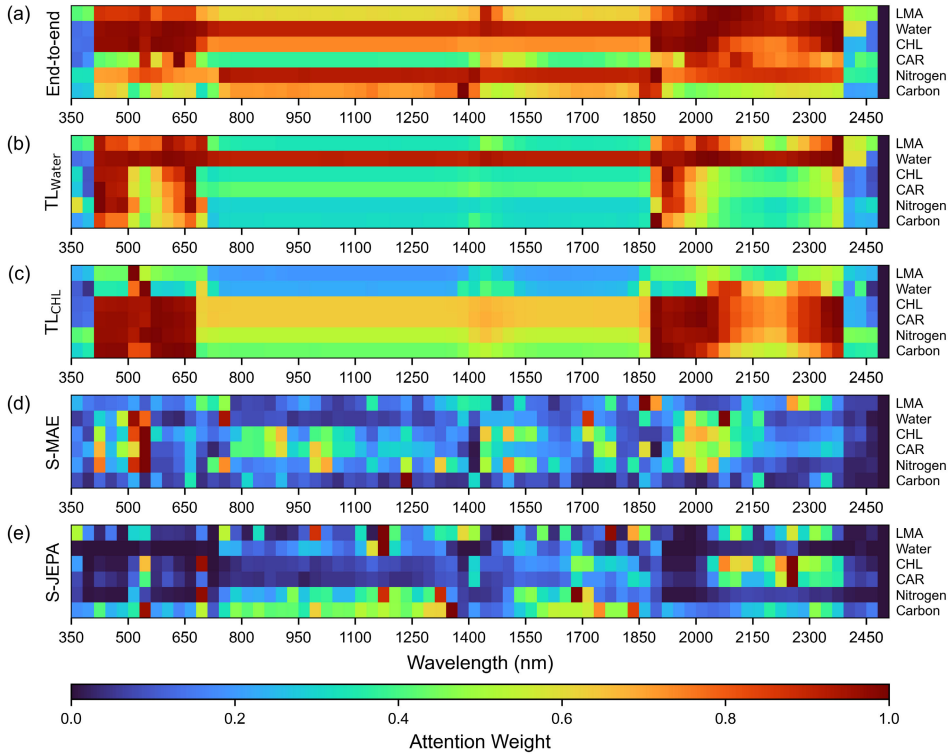


Fig. 8. Normalized attention weights across spectral bands for different models: (a) End-to-end, (b) TL_{Water}, (c) TL_{CHL}, (d) S-MAE, and (e) S-JEPA.

was notably negative. Correlation strength nd reliability exhibited a clear positive relationship across bootstrapped estimates, indicating that stronger trait associations were also estimated with higher confidence.

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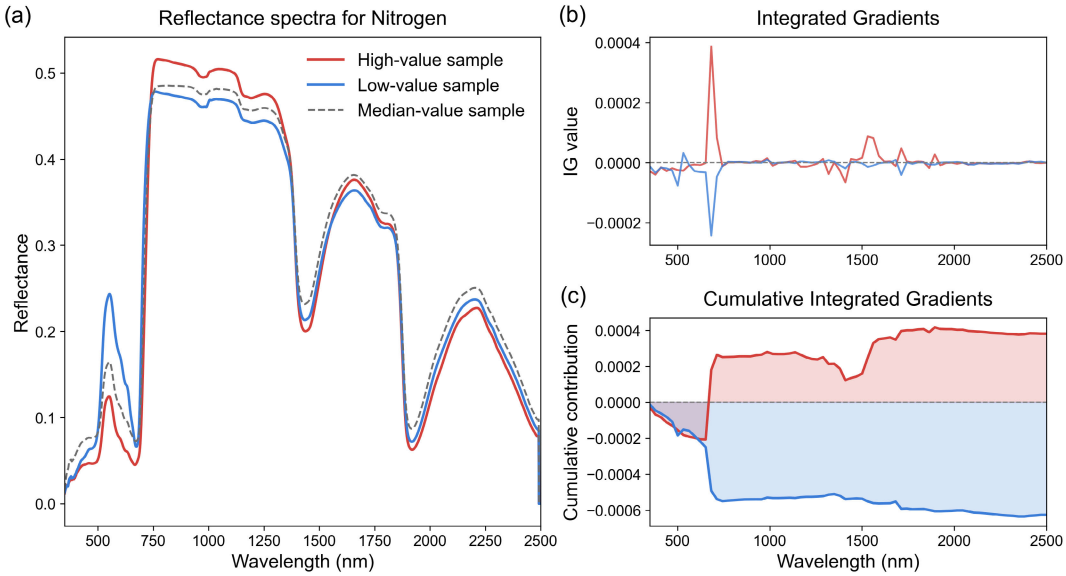


Fig. 9. IGs analysis for Nitrogen prediction. (a) Original reflectance spectra of high-, median-, and low-value samples. (b) Wavelength-wise IG contributions of high- and low-value samples relative to the median-value sample. (c) Corresponding cumulative IG contributions across the spectral range.

IV. Discussion

A. Generalizable Spectral Features Derived From SSL Boost Model Predictive Capacity

In spectrum–trait estimation, a primary limitation of the supervised learning paradigm is the scarcity of trait labels, which severely hinders the model's ability to identify informative spectral features from unlabeled samples. Accurately modeling the complex, nonlinear relationships between hyperspectral data and leaf traits requires a model with strong high-dimensional representation capabilities. Our results demonstrate the superior capacity of deep learning in this regard: End-to-end models consistently outperform PLSR when using the full set of labels (R^2 = 0.697 versus 0.655). This agrees with findings from previous studies [12], [38], [58]. However, supervised models exhibit significant limitations when trait labels are scarce. Because their optimization depends on minimizing prediction errors, missing labels prevent loss computation for a substantial portion of samples, leaving their underlying spectral features unused. When training a large-parameter deep model on only a small set of trait-labeled samples, the network tends to memorize noise or spurious patterns that are not representative of the overall data distribution, leading to poor generalization, a well-known manifestation of overfitting [59]. In our experiments, when only 5% of the labels were used, the R^2 of the End-to-end model dropped sharply to 0.121. MTL attempts to alleviate overfitting by learning shared representations across multiple traits, where the trade-off between multiple loss functions acts as a form of implicit regularization [60]. However, MTL inherently assumes concurrent availability of multiple trait labels, thereby failing to address the core challenge of label sparsity. Our results show that MTL significantly outperforms the End-to-end model when labels are relatively scarce (R^2 = 0.702 versus 0.411, at 100% proportion in densely labeled dataset), but its performance deteriorates rapidly with reduced label availability (R^2 = 0.146 at 10% proportion).

A further constraint of the supervised learning paradigm is the restricted scope of exploitable spectral features. As shown in Fig. 8(b) and (c), trait-specific supervised models, when transferred to predict other traits via TL, activate only subsets of highly similar spectral bands while disregarding others. This proves effective for aligned or correlated source-target traits; for instance, bidirectional CHL–CAR transfer yielded R^2 = 0.703 (100% proportion), and intra-trait transfer maintained robustness even at 1% labels (R^2 = 0.610). Similar findings have been reported in prior work [35], [36], [61], [62]. However, suboptimal alignment triggers negative transfer [41], yielding poor results despite ample labels (e.g., R^2 = 0.412 for CAR–Carbon at 100% proportion). This reveals a critical shortfall in supervised feature extraction, limiting adaptability across diverse tasks.

SSL paradigms, exemplified by S-JEPA and S-MAE, surmount these limitations by harnessing unlabeled data to learn generalizable spectral features. By pretraining on pretext tasks—latent space prediction for S-JEPA and masked reconstruction for S-MAE—these models exploit the entire dataset, irrespective of label availability, yielding robust performance under extreme scarcity (average R^2 > 0.5 with 5% labels). Moreover, SSL-derived features are disentangled from specific traits or environments, forming a universal pool that enhances multitrait estimation precision and robustness. When estimating various traits, SSL models flexibly select appropriate subsets from a universal pool of spectral features, as evidenced by their sparse and distinct attention patterns [see Fig. 8(d) and (e)]. These findings collectively suggest that SSL significantly reduces dependence on labeled data while simultaneously enhancing the precision and robustness of multitrait estimation.

Between the SSL variants, S-JEPA's joint-embedding approach confers subtle yet consistent advantages over

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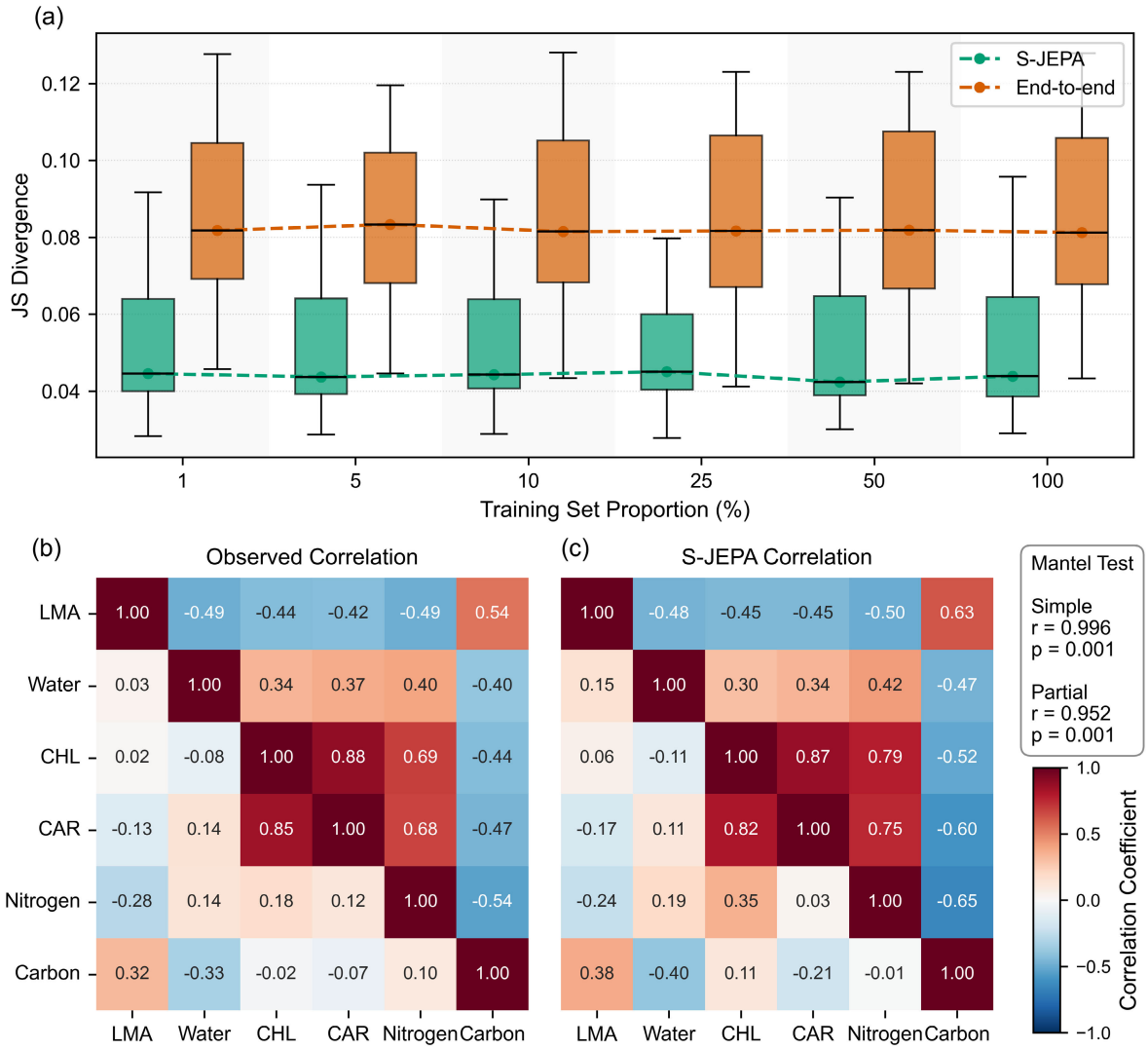


Fig. 10. (a) JS divergence between observed and predicted joint distributions by the S-JEPA and End-to-end models across training set proportions. (b) and (c) Trait correlation matrices from observations and S-JEPA predictions. In each subplot, the lower triangle shows simple correlations, and the upper triangle shows partial correlations.

S-MAE's reconstruction-based strategy. This is evidenced by our results, where S-JEPA generally outperformed S-MAE across traits (see Figs. 6 and 7), with further gains in area-unit experiments (R^2 = 0.805 versus R^2 = 0.793; Table F.1, Fig. F.1 in Appendix F of the supplementary material). In spite of its strengths in capturing spatial correlations via masking, S-MAE may emphasize high-covariance or finer-grained details susceptible to noise [63]. In contrast, S-JEPA's joint-embedding objective implicitly favors influential and predictive features while suppressing noisy signals, as supported by theoretical analyses of joint-embedding mechanisms [64]. This distinction enables S-JEPA's more flexible and abstract feature use, illustrating the methodological advances enabled by SSL in hyperspectral trait estimation.

B. Decoupled Application of Generalizable Spectral Features Significantly Amplifies Model Scalability in Practical Tasks

In this study, we primarily focused on predicting traits representing substance content using mass units (i.e., %DM) rather than area units (i.e., g/m^2). This choice was primarilydue to the generally larger errors associated with area measurements compared to mass measurements, and the difficulty in upholding the planar homogeneity assumption under realistic conditions [65]. Additionally, for substances with relatively stable content (e.g., Carbon), area-based representations can degenerate into scaled approximations of LMA, diminishing their meaning.

Nevertheless, the SSL model exhibited remarkable robustness when applied to trait estimation in area units. As shown in Appendix F of the supplementary material (Table F.1, Fig. F.1), S-JEPA (R^2 = 0.805) and S-MAE (R^2 = 0.793) both substantially exceed the performance of supervised models, including PLSR (R^2 = 0.698), End-to-end (R^2 = 0.705), and TL (R^2 = 0.690) models. These results underscore the scalability of the decoupled self-supervised paradigm, wherein the pretrained Embedder–Encoder modules extract generalizable spectral representations independent of any specific trait or measurement unit. Consequently, the learned features

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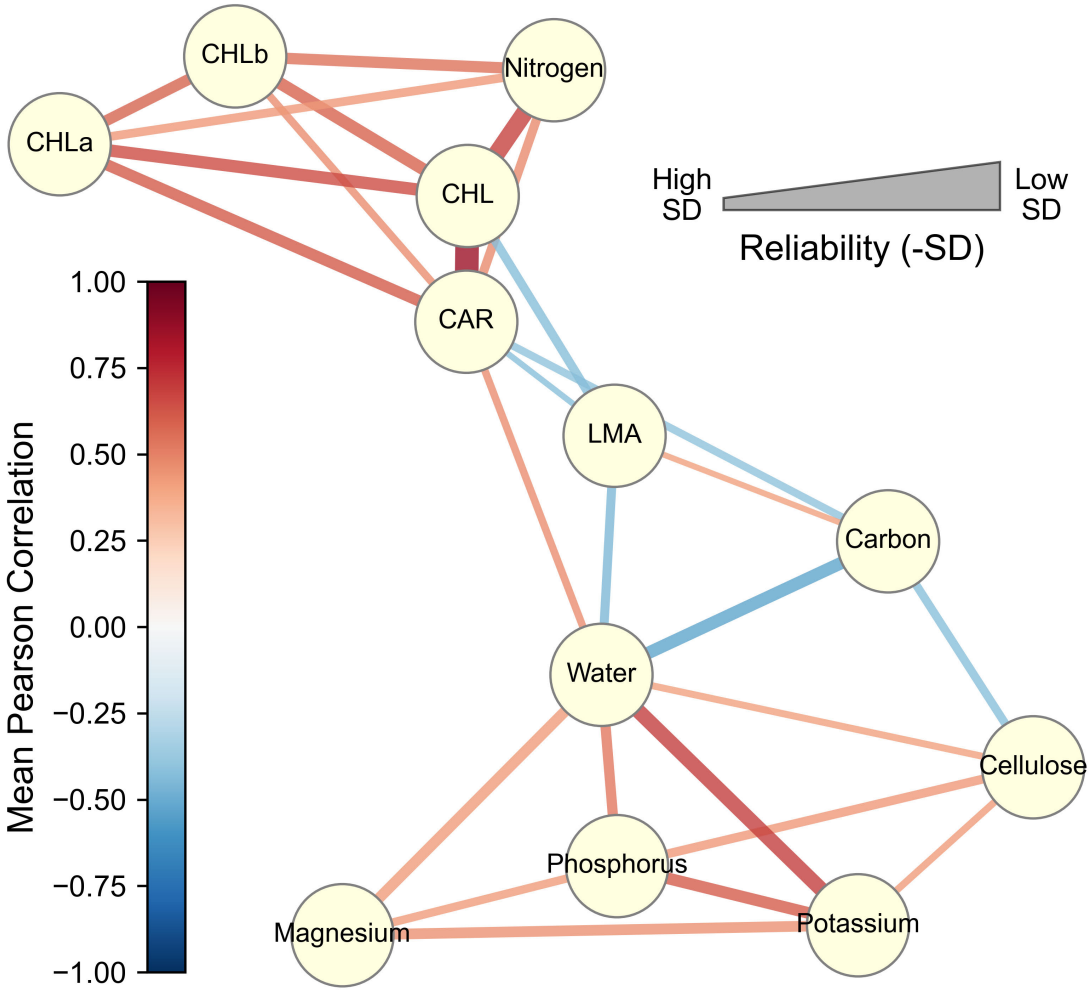


Fig. 11. Node network of trait correlations based on S-JEPA's predictions. Edges are color-coded by the magnitude and sign of the bootstrapped mean Pearson correlation. Edge width indicates the statistical reliability of the correlation, where thicker edges correspond to lower SD (higher reliability) and thinner edges correspond to higher SD (lower reliability).

TABLE III
TRAINING EFFICIENCY AND MODEL SIZE ACROSS LEARNING STAGES

Training Stage	Time per Epoch per Sample (ms)	Epochs to Convergence	Cumulative Time per Sample (ms)	Model Size (M parameters)
S-JEPA	0.495	1000	494.7	58.19
S-MAE	0.226	1000	226.3	58.19
Decoder-only	0.028	500	13.9	2.51
End-to-end	0.227	500	113.3	60.70
MTL (six traits)	0.272	5000	1360.1	73.25

retain compatibility across both mass-based and area-based formulations, enabling flexible downstream adaptation and significantly improving real-world applicability.

The decoupled architecture of the proposed framework systematically isolates generalizable feature learning from downstream trait estimation, yielding substantial dividends in both computational efficiency and deployment scalability. As summarized in Table III, although the SSL models involve a one-time pretraining investment, extending the model to new traits requires only the fine-tuning of a lightweight decoder. Compared to the End-to-end paradigm, this strategy reduces training time by 87.7% and parameter overhead by 95.9%. Furthermore, the SSL training paradigm demonstrates superior stability over MTL. MTL often suffers from unstable convergence due to gradient interference among multiple trait objectives, which was observed in both our experiments (see Fig. D.5 in Appendix D of the supplementary material) and prior literature [66], [67].

Beyond training efficiency, the shared-backbone architecture optimizes inference scalability for concurrent multitrait monitoring. Serial paradigms, such as End-to-end and TL, exhibit a linear computational complexity of $O(N)$ for N traits. In

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TABLE IV
COMPARISON OF S-JEPA WITH PREVIOUS HYPERSPECTRAL TRAIT ESTIMATION METHODS

Trait	Species	Model Type	R^2/nRMSE	Unit Type	Reference
Nitrogen	Grass	1D-CNN	0.72/0.140	Mass	[32]
	Grapevine	GPR	0.78/-	Mass	[65]
	Grass	Hybrid Model	0.78/-	Mass	[68]
	Integrated	S-JEPA	0.86/0.074	Mass	-
Water	Integrated	PLSR	0.82/0.062	Area	[69]
	Oilseed rape	CNN-LSTM-R	0.81/-	Mass	[70]
	Integrated	S-JEPA	0.95/0.025	Mass	-
	Integrated	S-JEPA	0.94/0.039	Area	-
LMA	Integrated	CNN (MTL)	0.69/0.107	Area	[38]
	Integrated	Ensemble on Random patches (TL)	≈0.75/-	Area	[61]
	Integrated	PLSR	≈0.70/0.093	Area	[71]
	Integrated	S-JEPA	0.71/0.068	Area	-

contrast, the SSL framework enables amortized inference with a complexity of $O(1 + \varepsilon N)$. In our implementation, ε was empirically determined to be 0.022. This indicates that the marginal cost of additional trait-specific decoders is negligible because the resource-intensive spectral encoding is performed only once. This "once-for-all" encoding mechanism allows S-JEPA to reconcile high predictive fidelity with extreme computational economy, making it uniquely qualified for real-time applications on resource-constrained platforms.

Our results confirm that the S-JEPA model either surpasses or matches the performance of antecedent methods (see Table IV). For nitrogen estimation, S-JEPA outperforms previous efforts employing convolutional neural networks or hybrid physical-empirical methods on grass or grapevine datasets [32], [65], [68]. In water content estimation, it achieves higher accuracy compared to earlier studies reliant on supervised machine learning techniques [69], [70]. For LMA estimation, our model provides performance commensurate with prior research [38], [61], [71], although it is based on a more expansive dataset. However, it is imperative to acknowledge that substantial differences exist across studies in terms of dataset curation, data preprocessing methodologies, and the metrics employed for evaluation. These discrepancies preclude a completely objective assessment of model superiority based solely on direct comparisons of predictive accuracy.

C. Integrated Multitrait Predictions Facilitate Insights Into Intricate Relationships From a Unified Spectral Perspective

Plant traits emerge from complex ecosystem interactions and trade-offs during growth [72], [73]. Traditionally, their relationships are analyzed via statistical methods on sparse experimental data [11], limited by bias, inaccuracies, and inconsistencies in design or sources. Hyperspectral techniques with machine learning have advanced trait retrieval [74], [75], [76]. Yet, these approaches primarily rely on supervised learning, where extracted features are biased toward labeled datasets and lack consistency across traits and studies. In contrast, the SSL models extract generalizable features from diverse hyperspectral datasets aggregated from multiple sources, allowing multitrait predictions based on a standardized feature set. This framework verifies transferability and reveals real-world multitrait dynamics beyond statistical metrics.

The attention weight maps and integrated-gradient analyses reveal how the models internally map spectral features to trait-specific predictions (see Figs. 8 and 9). Unlike the End-to-end model, which applies uniform or homogeneous spectral weighting, S-JEPA dynamically attends to trait-relevant bands, accurately identifying known sensitive regions for water, nitrogen, and CHL [32], [74], [77], [78]. This alignment between the model's spectral focus and established physiological mechanisms validates its interpretability and biological plausibility.

Quantitative analysis of statistics, as well as a holistic structure for multiple traits, further underscores S-JEPA's strength in capturing cross-trait dependencies (see Figs. 10 and 11). The model achieves a notably lower JS divergence from observed trait distributions than the supervised baseline (0.058 versus 0.093) and exhibits exceptional consistency across both simple and partial correlation matrices. The multitrait node network encapsulates ecophysiological interdependencies—water serving as a central hub [79]; pigment–nitrogen coupling supporting photosynthesis [4], [6], [80]; and potassium–water interplay regulating osmotic balance [81]. Crucially, these relationships are inferred without any explicit prior knowledge of trait synergy, showcasing S-JEPA's capability to uncover biologically meaningful linkages purely from hyperspectral data.

D. Limitations and Outlook

The proposed S-JEPA framework demonstrates strong capability in learning generalizable spectral features and predicting multiple traits from heterogeneous hyperspectral data. However, as the dataset is aggregated from diverse global sources, inevitable variations exist not only in species composition and ecological contexts but also in sensor characteristics and acquisition geometries. While our random subsampling strategy successfully validates the model's robustness on a global scale, explicitly quantifying and mitigating domain shifts across distinct biomes or plant functional types remains a valuable avenue for further refinement. Future work could explore unsupervised domain adaptation techniques to systematically

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align feature distributions, thereby ensuring high precision when transferring the model to specific, previously unseen regional contexts.

In predicting the multitrait matrix, a subset of traits relied on decoders trained with extremely limited labeled data, which inevitably introduces some deviations from real-world conditions. Despite these challenges, the multitrait network derived from bootstrapped analyses revealed highly consistent and biologically coherent inter-trait relationships. The stable recurrence of key physiological associations—such as the coupling between water and potassium, or the joint dynamics of nitrogen and CHL—demonstrates that the learned spectral representations preserve genuine ecological structures rather than statistical coincidences. This finding highlights the resilience of generalizable spectral features: even when trained on imperfect and heterogeneous data, they capture the underlying organization of plant functions with remarkable fidelity.

Building on this foundation, the flexible architecture of S-JEPA offers a versatile baseline for advancing ecosystem monitoring. The high-dimensional latent representations learned by S-JEPA are inherently compatible with multisource data fusion (e.g., integrating reflectance with LiDAR or thermal data). Moreover, by mitigating the reliance on extensive ground-truth labeling, this framework serves as a scalable engine for upcoming hyperspectral satellite missions (e.g., Surface Biology and Geology, SBG, and Copernicus Hyperspectral Imaging Mission, CHIME), enabling continuous, planetary-level observation. More fundamentally, emerging research suggests that spectral data may be regarded as a distinct trait, shaped by ecological or evolutionary pressures, rather than simply being a combination of other traits [12]. A self-supervised approach that prioritizes spectra over traditional trait definitions, therefore, holds transformative potential for understanding ecosystem dynamics. By focusing on spectral signatures, our study opens new research directions, potentially providing insights into environmental processes that go beyond traditional trait-based methods.

V. Conclusion

Contemporary hyperspectral trait estimation is constrained by its reliance on dense labeled datasets, whereas trait measurements are often sparse. To address this, we used a large leaf hyperspectral dataset and developed S-JEPA, an SSL framework. This approach enables the model to learn generalizable spectral features by predicting target region representations from context regions within spectral bands, thereby reducing the need for extensive trait labels during feature learning. Our results demonstrated that S-JEPA markedly outperformed conventional supervised, trait-label-dependent methods, and also surpassed S-MAE, another strong self-supervised baseline, by a small margin. Moreover, the S-JEPA-predicted multitrait relationships align well with observed patterns and underlying physiological mechanisms, providing a powerful tool for improving sparse datasets by inferring missing trait labels. With its low downstream training cost and adaptability to new traits and scenarios, this framework offers a scalable solution that holds significant promise for advancing ecosystem monitoring.

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References

[1] M. Migliavacca et al., "The three major axes of terrestrial ecosystem function," *Nature*, vol. 598, no. 7881, pp. 468–472, Oct. 2021, doi: 10.1038/s41586-021-03939-9.

[2] M. Dubbert, A. Kübert, and C. Werner, "Impact of leaf traits on temporal dynamics of transpired oxygen isotope signatures and its impact on atmospheric vapor," *Frontiers Plant Sci.*, vol. 8, Jan. 2017, Art. no. 5, doi: 10.3389/fpls.2017.00005.

[3] X. Feng and M. Dietze, "Scale dependence in the effects of leaf ecophysiological traits on photosynthesis: Bayesian parameterization of photosynthesis models," *New Phytol.*, vol. 200, no. 4, pp. 1132–1144, Dec. 2013, doi: 10.1111/nph.12454.

[4] B. Demmig-Adams and W. W. Adams, "The role of xanthophyll cycle carotenoids in the protection of photosynthesis," *Trends Plant Sci.*, vol. 1, no. 1, pp. 21–26, Jan. 1996, doi: 10.1016/s1360-1385(96)80019-7.

[5] W. Stahl and H. Sies, "Antioxidant activity of carotenoids," *Mol. Aspects Med.*, vol. 24, no. 6, pp. 345–351, Dec. 2003, doi: 10.1016/s0098-2997(03)00030-x.

[6] H. Croft, J. M. Chen, X. Luo, P. Bartlett, B. Chen, and R. M. Staebler, "Leaf chlorophyll content as a proxy for leaf photosynthetic capacity," *Global Change Biol.*, vol. 23, no. 9, pp. 3513–3524, Sep. 2017, doi: 10.1111/gcb.13599.

[7] J. L. D. Osnas, J. W. Lichstein, P. B. Reich, and S. W. Pacala, "Global leaf trait relationships: Mass, area, and the leaf economics spectrum," *Science*, vol. 340, no. 6133, pp. 741–744, May 2013, doi: 10.1126/science.1231574.

[8] Z. Wang et al., "Leaf water content contributes to global leaf trait relationships," *Nature Commun.*, vol. 13, no. 1, p. 5525, Sep. 2022, doi: 10.1038/s41467-022-32784-1.

[9] A. Chlus and P. A. Townsend, "Characterizing seasonal variation in foliar biochemistry with airborne imaging spectroscopy," *Remote Sens. Environ.*, vol. 275, Jun. 2022, Art. no. 113023, doi: 10.1016/j.rse.2022.113023.

[10] E. Cui, E. Weng, E. Yan, and J. Xia, "Robust leaf trait relationships across species under global environmental changes," *Nature Commun.*, vol. 11, no. 1, p. 2999, Jun. 2020, doi: 10.1038/s41467-020-16839-9.

[11] J. L. Funk et al., "Revisiting the holy grail: Using plant functional traits to understand ecological processes," *Biol. Rev.*, vol. 92, no. 2, pp. 1156–1173, May 2017, doi: 10.1111/brv.12275.

[12] S. Kothari and A. K. Schweiger, "Plant spectra as integrative measures of plant phenotypes," *J. Ecology*, vol. 110, no. 11, pp. 2536–2554, Nov. 2022, doi: 10.1111/1365-2745.13972.

[13] J. A. Prananto, B. Minasny, and T. Weaver, "Chapter one–near infrared (NIR) spectroscopy as a rapid and cost-effective method for nutrient analysis of plant leaf tissues," in *Advances in Agronomy*, vol. 164. New York, NY, USA: Academic, 2020, pp. 1–49, doi: 10.1016/bs.agron.2020.06.001.

[14] P. J. Curran, "Remote sensing of foliar chemistry," *Remote Sens. Environ.*, vol. 30, no. 3, pp. 271–278, Dec. 1989, doi: 10.1016/0034-4257(89)90069-2.

[15] R. F. Kokaly, "Investigating a physical basis for spectroscopic estimates of leaf nitrogen concentration," *Remote Sens. Environ.*, vol. 75, no. 2, pp. 153–161, Feb. 2001, doi: 10.1016/s0034-4257(00)00163-2.

[16] S. Jacquemoud and F. Baret, "PROSPECT: A model of leaf optical properties spectra," *Remote Sens. Environ.*, vol. 34, no. 2, pp. 75–91, Nov. 1990, doi: 10.1016/0034-4257(90)90100-z.

[17] J.-B. Feret et al., "PROSPECT-4 and 5: Advances in the leaf optical properties model separating photosynthetic pigments," *Remote Sens. Environ.*, vol. 112, no. 6, pp. 3030–3043, Jun. 2008, doi: 10.1016/j.rse.2008.02.012.

[18] J.-B. Féret, A. A. Gitelson, S. D. Noble, and S. Jacquemoud, "PROSPECT-D: Towards modeling leaf optical properties through a complete lifecycle," *Remote Sens. Environ.*, vol. 193, pp. 204–215, May 2017, doi: 10.1016/j.rse.2017.03.004.

[19] J. Jiang, A. Comar, M. Weiss, and F. Baret, "FASPECT: A model of leaf optical properties accounting for the differences between upper and lower faces," *Remote Sens. Environ.*, vol. 253, Feb. 2021, Art. no. 112205, doi: 10.1016/j.rse.2020.112205.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 14:48:07 UTC from IEEE Xplore. Restrictions apply.

[20] F. Yu et al., "PIOSL: An improved version of PROSPECT model for simulating leaf optical properties by considering leaf internal structural variations," *Comput. Electron. Agricult.*, vol. 211, Aug. 2023, Art. no. 108030, doi: 10.1016/j.compag.2023.108030.

[21] R. Darvishzadeh, A. Skidmore, M. Schlerf, and C. Atzberger, "Inversion of a radiative transfer model for estimating vegetation LAI and chlorophyll in a heterogeneous grassland," *Remote Sens. Environ.*, vol. 112, no. 5, pp. 2592–2604, May 2008, doi: 10.1016/j.rse.2007.12.003.

[22] B. Lu, C. Proctor, and Y. He, "Investigating different versions of PROSPECT and PROSAIL for estimating spectral and biophysical properties of photosynthetic and non-photosynthetic vegetation in mixed grasslands," *GIScience Remote Sens.*, vol. 58, no. 3, pp. 354–371, Apr. 2021, doi: 10.1080/15481603.2021.1877435.

[23] M. Liu, Z. Sun, S. Lu, and K. Omasa, "Combining multiangular, polarimetric, and hyperspectral measurements to estimate leaf nitrogen concentration from different plant species," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 4404715, doi: 10.1109/TGRS.2021.3106786.

[24] R.-H. Shi, D.-F. Zhuang, and Z. Niu, "Simulated hyperspectral data analysis using continuum removal: Case study on leaf chlorophyll prediction," in *Proc. 8th Int. Conf. Signal Process.*, Nov. 2006, doi: 10.1109/icosp.2006.345949.

[25] F. Tsai and W. Philpot, "Derivative analysis of hyperspectral data," *Remote Sens. Environ.*, vol. 66, no. 1, pp. 41–51, Oct. 1998, doi: 10.1016/s0034-4257(98)00032-7.

[26] W. Sun and Q. Du, "Hyperspectral band selection: A review," *IEEE Geosci. Remote Sens. Mag.*, vol. 7, no. 2, pp. 118–139, Jun. 2019, doi: 10.1109/MGRS.2019.2911100.

[27] H. Chen, H. Li, Z. Liu, C. Zhang, S. Zhang, and P. M. Atkinson, "A novel greenness and water content composite index (GWCCI) for soybean mapping from single remotely sensed multispectral images," *Remote Sens. Environ.*, vol. 295, Sep. 2023, Art. no. 113679, doi: 10.1016/j.rse.2023.113679.

[28] B.-C. Gao, "NDWI—A normalized difference water index for remote sensing of vegetation liquid water from space," *Remote Sens. Environ.*, vol. 58, no. 3, pp. 257–266, Dec. 1996, doi: 10.1016/s0034-4257(96)00067-3.

[29] C. Ordóñez, J. R. Rodríguez-Pérez, J. J. Moreira, and E. Sanz, "Using hyperspectral spectrometry and functional models to characterize vine-leaf composition," *IEEE Trans. Geosci. Remote Sens.*, vol. 51, no. 5, pp. 2610–2618, May 2013, doi: 10.1109/TGRS.2012.2217344.

[30] S. Deng et al., "Development of a nondestructive prediction model for nutrient content of orchid leaves using different hyperspectral sensors," *Comput. Electron. Agricult.*, vol. 237, Oct. 2025, Art. no. 110519, doi: 10.1016/j.compag.2025.110519.

[31] D. Heckmann, U. Schlüter, and A. P. M. Weber, "Machine learning techniques for predicting crop photosynthetic capacity from leaf reflectance spectra," *Mol. Plant*, vol. 10, no. 6, pp. 878–890, Jun. 2017, doi: 10.1016/j.molp.2017.04.009.

[32] R. R. Pullanagari, M. Dehghan-Shoar, I. J. Yule, and N. Bhatia, "Field spectroscopy of canopy nitrogen concentration in temperate grasslands using a convolutional neural network," *Remote Sens. Environ.*, vol. 257, May 2021, Art. no. 112353, doi: 10.1016/j.rse.2021.112353.

[33] S. Yu et al., "Deep learning models based on hyperspectral data and time-series phenotypes for predicting quality attributes in lettuces under water stress," *Comput. Electron. Agricult.*, vol. 211, Aug. 2023, Art. no. 108034, doi: 10.1016/j.compag.2023.108034.

[34] Y. Ma, S. Chen, S. Ermon, and D. B. Lobell, "Transfer learning in environmental remote sensing," *Remote Sens. Environ.*, vol. 301, Feb. 2024, Art. no. 113924, doi: 10.1016/j.rse.2023.113924.

[35] F. Ji et al., "Leveraging transfer learning and leaf spectroscopy for leaf trait prediction with broad spatial, species, and temporal applicability," *Remote Sens. Environ.*, vol. 326, Aug. 2025, Art. no. 114818, doi: 10.1016/j.rse.2025.114818.

[36] L. Wan, W. Zhou, Y. He, T. C. Wanger, and H. Cen, "Combining transfer learning and hyperspectral reflectance analysis to assess leaf nitrogen concentration across different plant species datasets," *Remote Sens. Environ.*, vol. 269, Feb. 2022, Art. no. 112826, doi: 10.1016/j.rse.2021.112826.

[37] J. Yue et al., "Estimating crop leaf area index and chlorophyll content using a deep learning-based hyperspectral analysis method," *Comput. Electron. Agricult.*, vol. 227, Dec. 2024, Art. no. 109653, doi: 10.1016/j.compag.2024.109653.

[38] E. Cherif et al., "From spectra to plant functional traits: Transferable multi-trait models from heterogeneous and sparse data," *Remote Sens. Environ.*, vol. 292, Jul. 2023, Art. no. 113580, doi: 10.1016/j.rse.2023.113580.

[39] P. Cunningham, M. Cord, and S. J. Delany, "Supervised learning," in *Machine Learning Techniques for Multimedia: Case Studies on Organization and Retrieval*, M. Cord and P. Cunningham, Eds., Berlin, Germany: Springer, 2008, pp. 21–49, doi: 10.1007/978-3-540-75171-7_2.

[40] S. J. Pan and Q. Yang, "A survey on transfer learning," *IEEE Trans. Knowl. Data Eng.*, vol. 22, no. 10, pp. 1345–1359, Oct. 2010, doi: 10.1109/TKDE.2009.191.

[41] W. Zhang, L. Deng, L. Zhang, and D. Wu, "A survey on negative transfer," *IEEE/CAA J. Autom. Sinica*, vol. 10, no. 2, pp. 305–329, Feb. 2023, doi: 10.1109/JAS.2022.106004.

[42] E. P. Wagner, J. Merz, and P. A. Townsend, "Ecological spectral information system: An open spectral library," in *Proc. AGU Fall Meeting Abstracts*, Dec. 2018, pp. 1–4.

[43] J. Gui et al., "A survey on self-supervised learning: Algorithms, applications, and future trends," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 46, no. 12, pp. 9052–9071, Dec. 2024, doi: 10.1109/TPAMI.2024.3415112.

[44] M. Assran et al., "Self-supervised learning from images with a joint-embedding predictive architecture," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2023, doi: 10.1109/CVPR52729.2023.01499.

[45] K. He, X. Chen, S. Xie, Y. Li, P. Dollár, and R. Girshick, "Masked autoencoders are scalable vision learners," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2022. Accessed: Mar. 27, 2025. [Online]. Available: https://openaccess.thecvf.com/content/CVPR2022/html/He_Masked_Autoencoders_Are_Scalable_Vision_Learners_CVPR_2022_paper.html

[46] J. Devlin, M.-W. Chang, K. Lee, and K. Toutanova, "BERT: Pre-training of deep bidirectional transformers for language understanding," in *Proc. Conf. North Amer. Chapter Assoc. Comput. Linguistics, Hum. Lang. Technol.*, vol. 1, Jun. 2019, pp. 4171–4186, doi: 10.18653/v1/N19-1423.

[47] A. Radford, J. Wu, R. Child, D. Luan, D. Amodei, and I. Sutskever. (2019). *Language Models Are Unsupervised Multitask Learners*. Accessed: Nov. 27, 2024. [Online]. Available: https://www.semanticscholar.org/paper/Language-Models-are-Unsupervised-Multitask-Learners-Radford-Wu/9405cc0d6169988371b2755e573cc28650d14dfe

[48] L. Bai, X. Zhang, H. Wang, and S. Du, "Integrating remote sensing with OpenStreetMap data for comprehensive scene understanding through multi-modal self-supervised learning," *Remote Sens. Environ.*, vol. 318, Mar. 2025, Art. no. 114573, doi: 10.1016/j.rse.2024.114573.

[49] X. Chen, Y. Zhang, Y. Dong, and B. Du, "Generative self-supervised learning with spectral–spatial masking for hyperspectral target detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5522713, doi: 10.1109/TGRS.2024.3423781.

[50] W. Liu et al., "Self-supervised feature learning based on spectral masking for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 4407715, doi: 10.1109/TGRS.2023.3310489.

[51] R. W. Schafer, "What is a Savitzky–Golay filter? [Lecture notes]," *IEEE Signal Process. Mag.*, vol. 28, no. 4, pp. 111–117, Jul. 2011, doi: 10.1109/MSP.2011.941097.

[52] S. C. Hicks, K. Okrah, J. N. Paulson, J. Quackenbush, R. A. Irizarry, and H. C. Bravo, "Smooth quantile normalization," *Biostatistics*, vol. 19, no. 2, pp. 185–198, Apr. 2018, doi: 10.1093/biostatistics/kxx028.

[53] A. Dosovitskiy et al., "An image is worth 16×16 words: Transformers for image recognition at scale," *presented at the Int. Conf. Learn. Represent.*, Oct. 2020. [Online]. Available: https://openreview.net/forum?id=YicbFdNTTy

[54] A. Vaswani et al., "Attention is all you need," in *Proc. Int. Conf. Neural Inf. Process. Syst. (NIPS)*, vol. 2017, pp. 6000–6010.

[55] D. Hong et al., "SpectralFormer: Rethinking hyperspectral image classification with transformers," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5518615, doi: 10.1109/TGRS.2021.3130716.

[56] S. Wold, M. Sjöström, and L. Eriksson, "PLS-regression: A basic tool of chemometrics," *Chemometric Intell. Lab. Syst.*, vol. 58, no. 2, pp. 109–130, Oct. 2001, doi: 10.1016/s0169-7439(01)00155-1.

[57] M. Sundararajan, A. Taly, and Q. Yan, "Axiomatic attribution for deep networks," 2017, *arXiv:1703.01365*.

[58] X. Deng et al., "Estimation of photosynthetic parameters from hyperspectral images using optimal deep learning architecture," *Comput. Electron. Agricult.*, vol. 216, Jan. 2024, Art. no. 108540, doi: 10.1016/j.compag.2023.108540.

[59] S. Salman and X. Liu, "Overfitting mechanism and avoidance in deep neural networks," 2019, *arXiv:1901.06566*.

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[60] S. Ruder, "An overview of multi-task learning in deep neural networks," 2017, *arXiv:1706.05098*.

[61] S. Fei et al., "Improved random patches and model transfer for deriving leaf mass per area across multispecies from spectral reflectance," *Comput. Electron. Agricult.*, vol. 218, Mar. 2024, Art. no. 108745, doi: 10.1016/j.compag.2024.108745.

[62] Y. Zhang et al., "Transfer-learning-based approach for leaf chlorophyll content estimation of winter wheat from hyperspectral data," *Remote Sens. Environ.*, vol. 267, Dec. 2021, Art. no. 112724, doi: 10.1016/j.rse.2021.112724.

[63] H. Van Assel, M. Ibrahim, T. Biancalani, A. Regev, and R. Balestriero, "Joint embedding vs reconstruction: Provable benefits of latent space prediction for self supervised learning," 2025, *arXiv:2505.12477*.

[64] E. Littwin et al., "How JEPA avoids noisy features: The implicit bias of deep linear self distillation networks," in *Proc. Adv. Neural Inf. Process. Syst.*, Dec. 2024, pp. 91300–91336.

[65] S. Peanusaha, A. Pourreza, Y. Kamiya, M. W. Fidelibus, and M. Chakraborty, "Nitrogen retrieval in grapevine (Vitis vinifera L.) leaves by hyperspectral sensing," *Remote Sens. Environ.*, vol. 302, Mar. 2024, Art. no. 113966, doi: 10.1016/j.rse.2023.113966.

[66] R. Cipolla, Y. Gal, and A. Kendall, "Multi-task learning using uncertainty to weigh losses for scene geometry and semantics," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, Jun. 2018, pp. 7482–7491, doi: 10.1109/CVPR.2018.00781.

[67] B. Lin et al., "Dual-balancing for multi-task learning," 2023, *arXiv:2308.12029*.

[68] M. H. Dehghan-Shoar, A. A. Orsi, R. R. Pullanagari, and I. J. Yule, "A hybrid model to predict nitrogen concentration in heterogeneous grassland using field spectroscopy," *Remote Sens. Environ.*, vol. 285, Feb. 2023, Art. no. 113385, doi: 10.1016/j.rse.2022.113385.

[69] F. Ji et al., "Unveiling the transferability of PLSR models for leaf trait estimation: Lessons from a comprehensive analysis with a novel global dataset," *New Phytol.*, vol. 243, no. 1, pp. 111–131, May 2024, doi: 10.1111/nph.19807.

[70] C. Zhang et al., "Leaf water content determination of oilseed rape using near-infrared hyperspectral imaging with deep learning regression methods," *Infr. Phys. Technol.*, vol. 134, Nov. 2023, Art. no. 104921, doi: 10.1016/j.infrared.2023.104921.

[71] Z. Wang et al., "Generality of leaf spectroscopic models for predicting key foliar functional traits across continents: A comparison between physically- and empirically-based approaches," *Remote Sens. Environ.*, vol. 293, Aug. 2023, Art. no. 113614, doi: 10.1016/j.rse.2023.113614.

[72] S. Díaz et al., "The global spectrum of plant form and function," *Nature*, vol. 529, no. 7585, pp. 167–171, Jan. 2016, doi: 10.1038/nature16489.

[73] Z. Wang, P. A. Townsend, and E. L. Kruger, "Leaf spectroscopy reveals divergent inter-and intra-species foliar trait covariation and trait–environment relationships across NEON domains," *New Phytologist*, vol. 235, no. 3, pp. 923–938, Aug. 2022, doi: 10.1111/nph.18204.

[74] S. Junttila et al., "Close-range hyperspectral spectroscopy reveals leaf water content dynamics," *Remote Sens. Environ.*, vol. 277, Aug. 2022, Art. no. 113071, doi: 10.1016/j.rse.2022.113071.

[75] S. P. Serbin, A. Singh, B. E. McNeil, C. C. Kingdon, and P. A. Townsend, "Spectroscopic determination of leaf morphological and biochemical traits for northern temperate and boreal tree species," *Ecological Appl.*, vol. 24, no. 7, pp. 1651–1669, Oct. 2014, doi: 10.1890/13-2110.1.

[76] N. K. Wijewardane, H. Zhang, J. Yang, J. C. Schnable, D. P. Schachtman, and Y. Ge, "A leaf-level spectral library to support high-throughput plant phenotyping: Predictive accuracy and model transfer," *J. Experim. Botany*, vol. 74, no. 14, pp. 4050–4062, Aug. 2023, doi: 10.1093/jxb/erad129.

[77] Y. Tang, F. Li, Y. Hu, and K. Yu, "Exploring the optimal wavelet function and wavelet feature for estimating maize leaf chlorophyll content," *IEEE Trans. Geosci. Remote Sens.*, vol. 63, 2024, Art. no. 4400812, doi: 10.1109/TGRS.2024.3513233.

[78] Y. Xiao, W. Zhao, D. Zhou, and H. Gong, "Sensitivity analysis of vegetation reflectance to biochemical and biophysical variables at leaf, canopy, and regional scales," *IEEE Trans. Geosci. Remote Sens.*, vol. 52, no. 7, pp. 4014–4024, Jul. 2014, doi: 10.1109/TGRS.2013.2278838.

[79] M. Fang, W. Ju, W. Zhan, T. Cheng, F. Qiu, and J. Wang, "A new spectral similarity water index for the estimation of leaf water content from hyperspectral data of leaves," *Remote Sens. Environ.*, vol. 196, pp. 13–27, Jul. 2017, doi: 10.1016/j.rse.2017.04.029.

[80] Y. Fu et al., "An overview of crop nitrogen status assessment using hyperspectral remote sensing: Current status and perspectives," *Eur. J. Agronomy*, vol. 124, Mar. 2021, Art. no. 126241, doi: 10.1016/j.eja.2021.126241.

[81] W. Hu et al., "The reduction in leaf area precedes that in photosynthesis under potassium deficiency: The importance of leaf anatomy," *New Phytol.*, vol. 227, no. 6, pp. 1749–1763, Sep. 2020, doi: 10.1111/nph.16644.

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